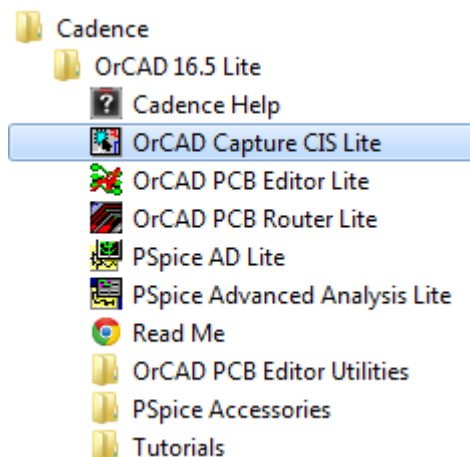


EBN 111 Practical 1 Tutorial

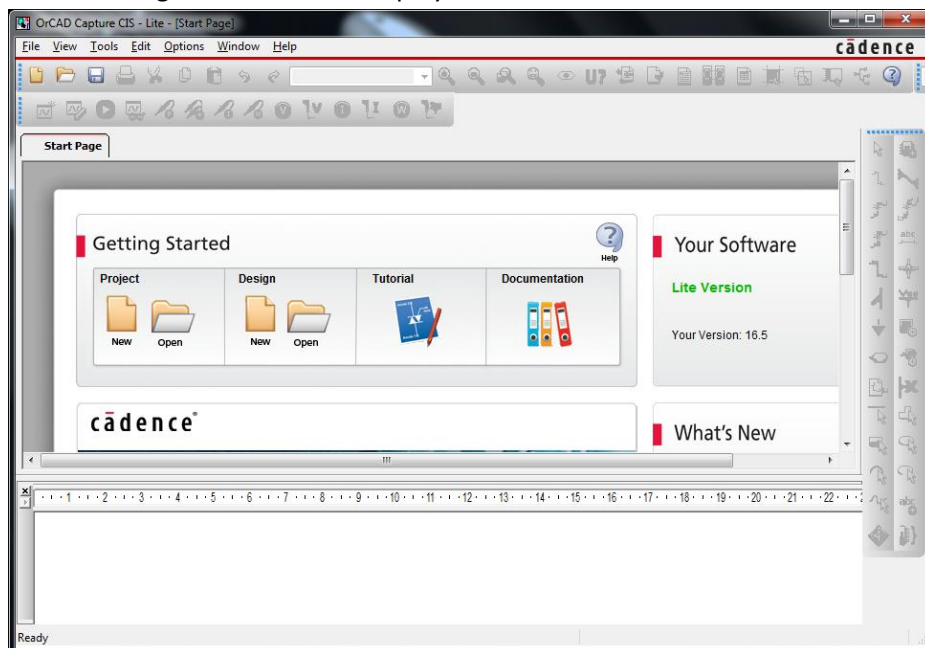
This tutorial serves as a guideline to installing and using the OrCAD Capture CIS and associated PSpice software.

1. Obtain the OrCAD Demo package. It can be downloaded from the following link:
ftp://ftp.ee.up.ac.za/pub/windows/eda/OrCAD_16_5_Full_Demo.zip
It is highly recommended that you download the software from a computer on campus, as the file is approximately 830 MB in size.
2. Extract the archive, and install the software using *setup.exe* which can be located in the *Disk 1* directory. This will take some time.
3. The following directory should have appeared in your Start menu:



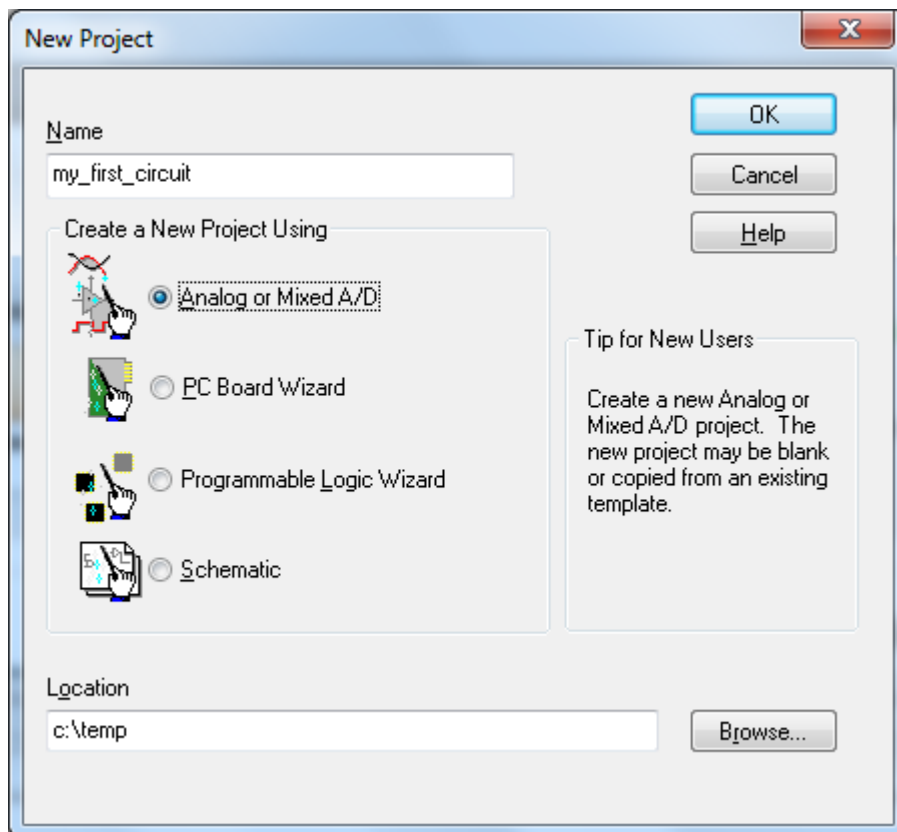
Launch the software by clicking on OrCAD Capture CIS Lite as highlighted above.

4. The following window will be displayed:

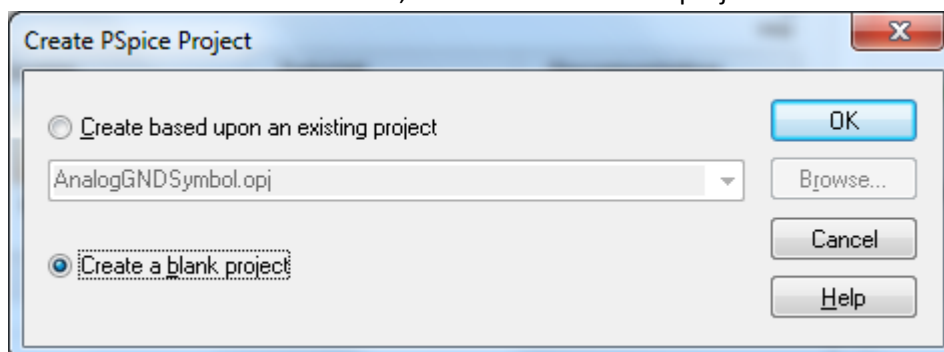


Create a new document either by clicking on *File -> New -> Project* or on the icon in the top-left of the screen.

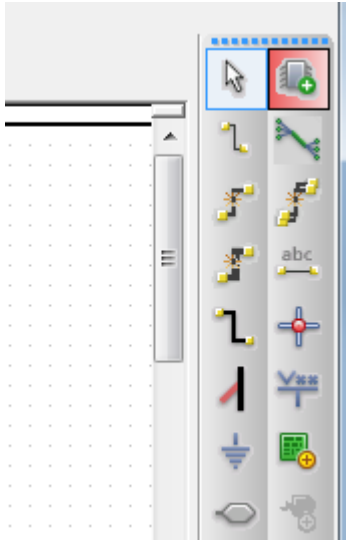
5. In the subsequent window, give your circuit a name, assign a location for its files to be saved, and select "Analog or Mixed A/D" as its type. Then click OK.



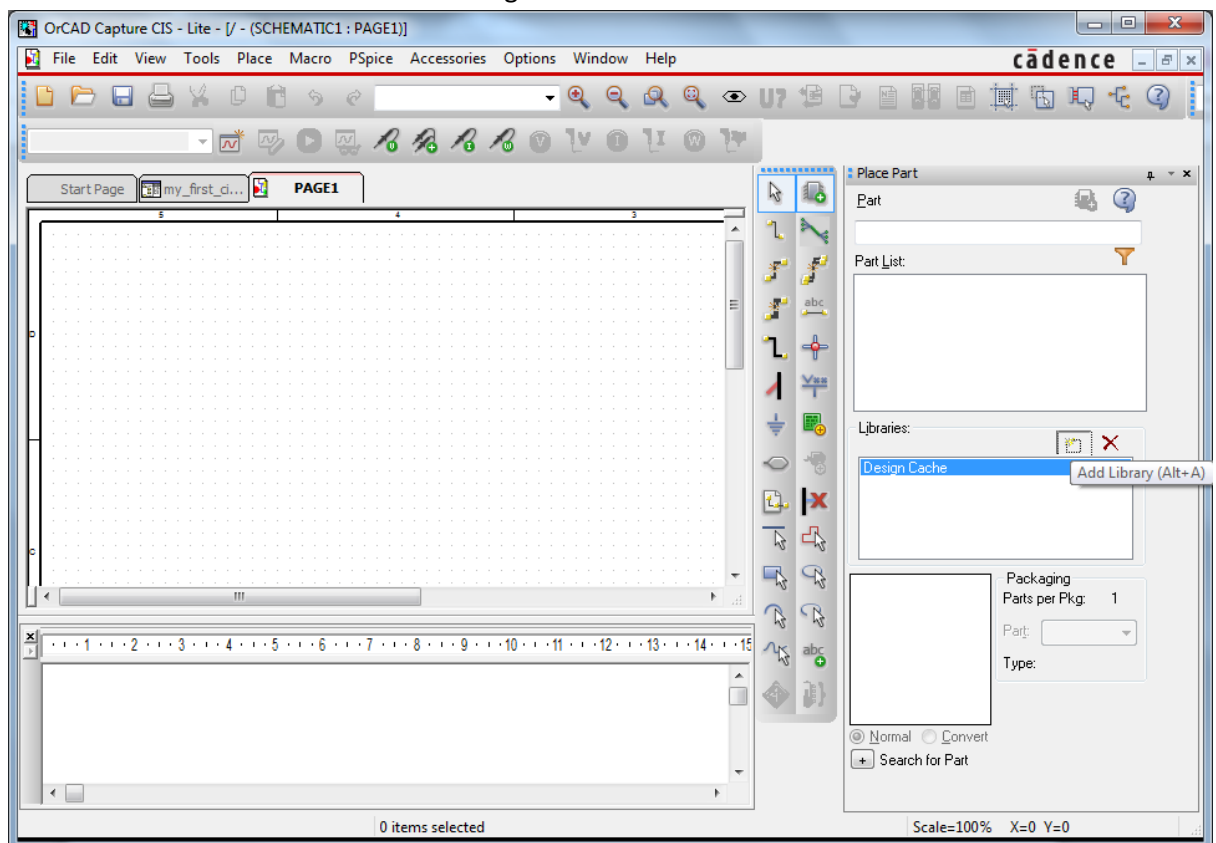
In the window which follows this, select “Create a blank project” and click OK.



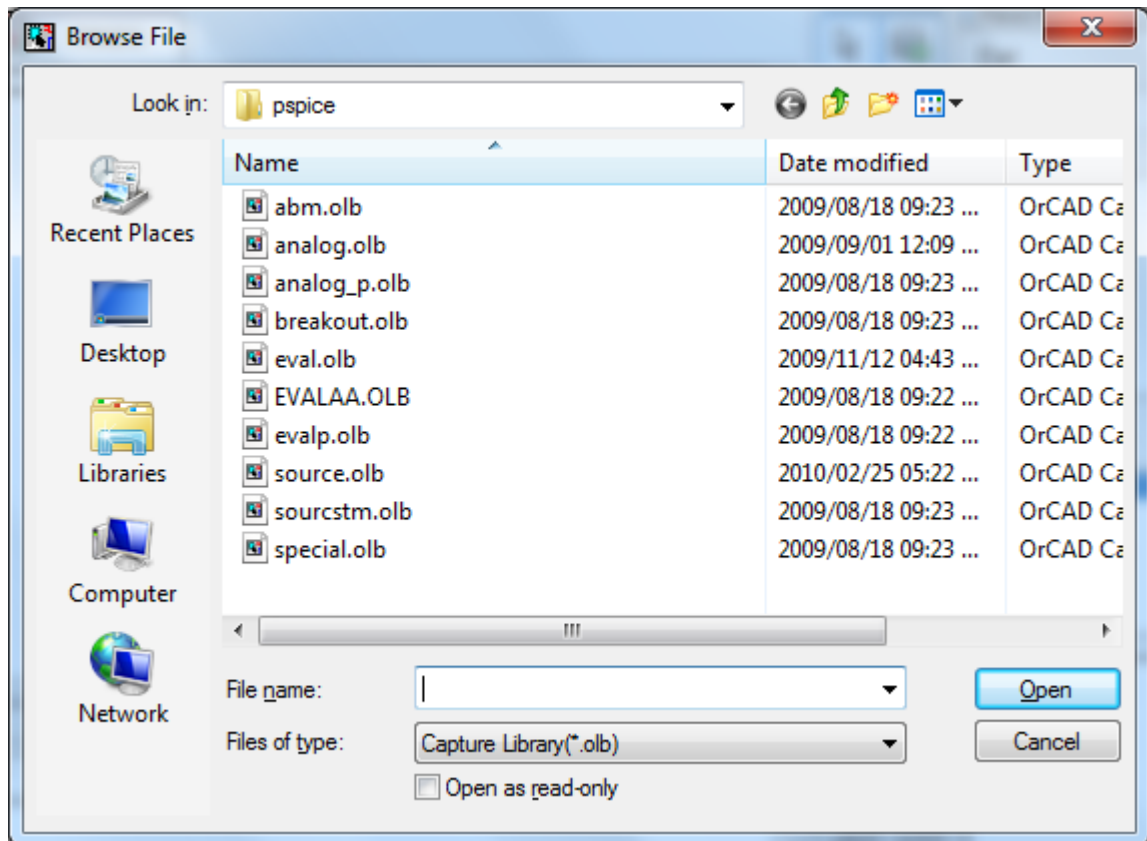
6. You will then be presented with a blank schematic. Before you can place components on your schematic, OrCAD needs libraries from which to take these components. Click on the “Place part” icon on the top right of the screen.



A window will appear on the right hand side of the screen. In it, click on “Add library” which can be found towards the middle of the right-hand side.



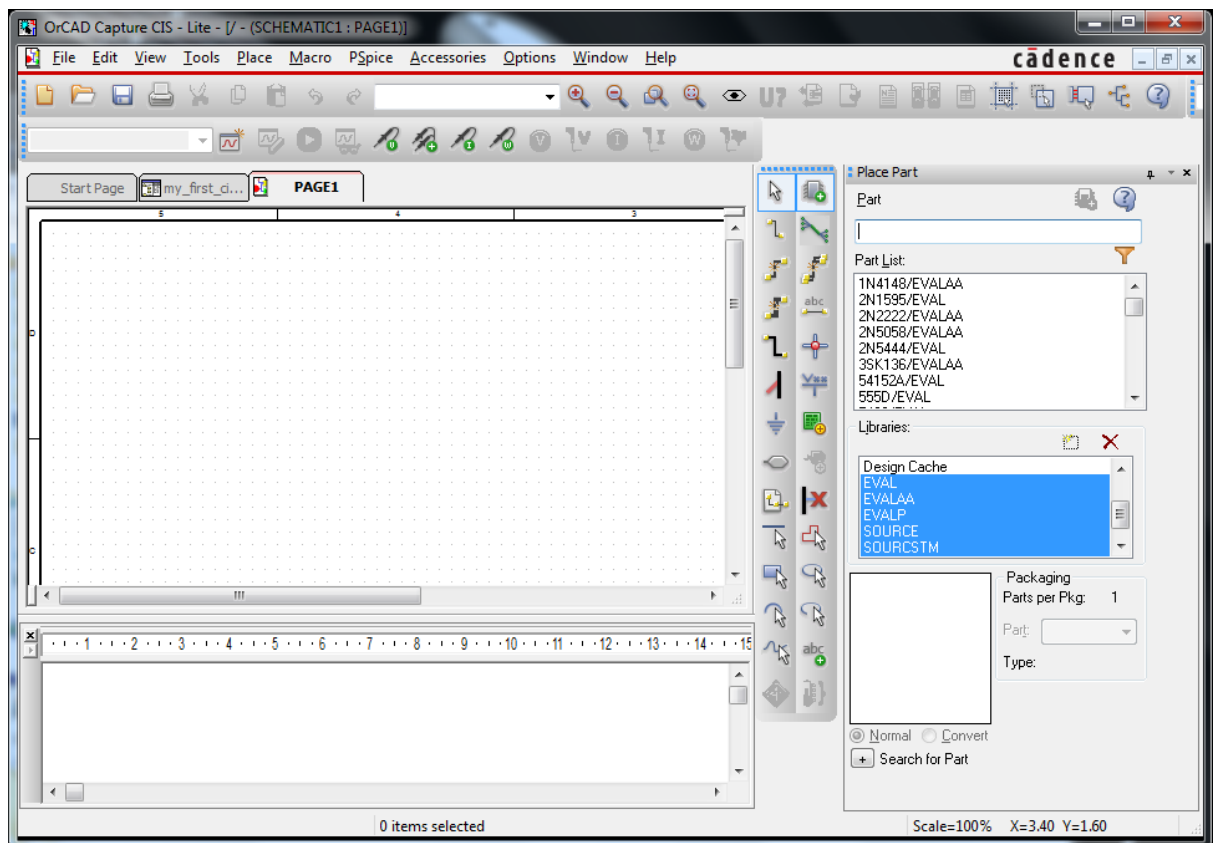
7. A window should open, showing you the contents of a directory named “pspice”. In the event that this does not happen, please navigate to
C:\OrCAD\OrCAD_16.5_Lite\tools\capture\library\pspice
or similar location in your install directory, so that you see the following library files:



If you do not import the correct library files, PSpice will not be able to simulate your circuit correctly!

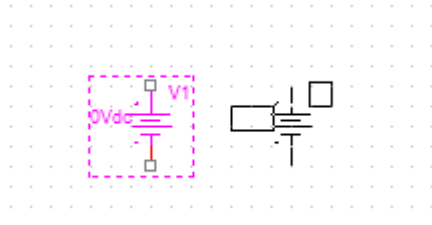
Select all the files, then click OK.

8. The boxes on the right-hand side of your screen will now be populated with names of circuit components:



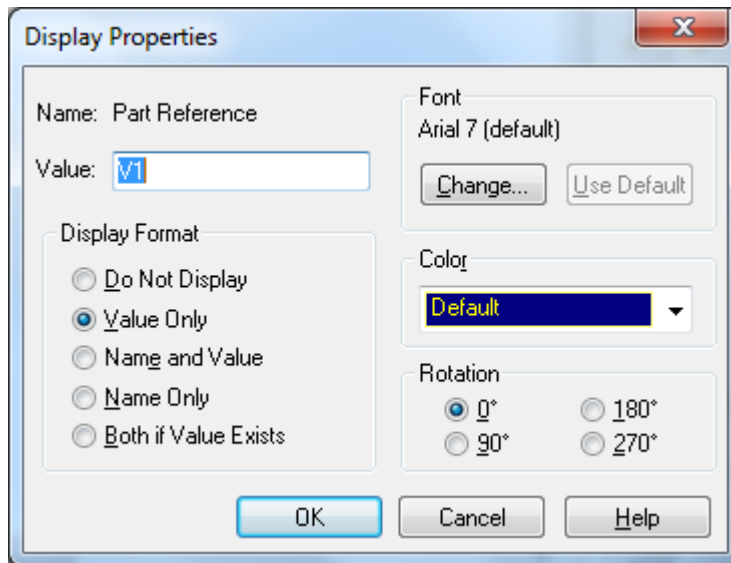
You are now ready to begin building a circuit schematic.

9. We will construct and simulate a simple circuit similar to that required in the pre-prac assignment. In the box labelled "Part" type "vdc" in order to place a DC voltage source. An item labelled "VDC/SOURCE" will be highlighted. Double-click on it and place the DC voltage source somewhere on your schematic.



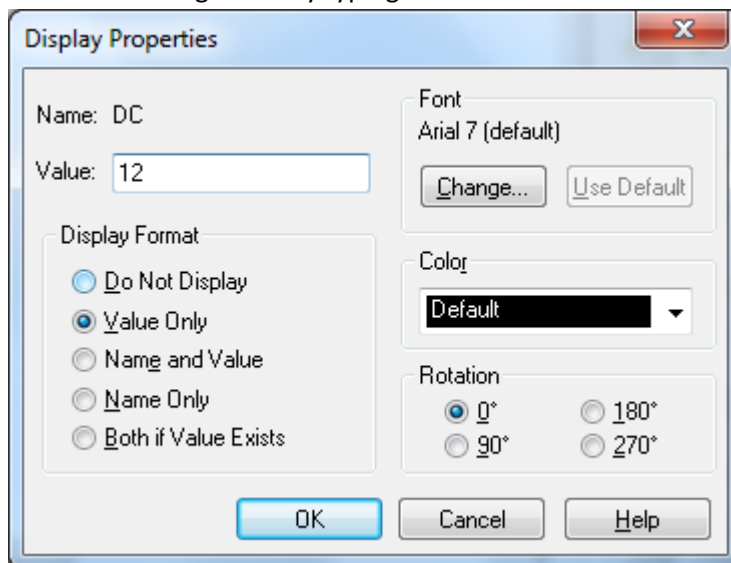
Press Esc. to return your cursor to normal.

10. Double-click on the label “V1”



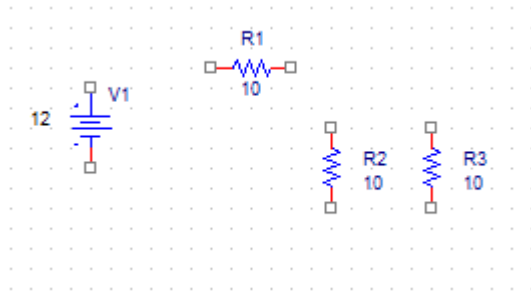
Should you desire (and this is often an advantage in more complicated circuits) you may rename your voltage source. If you rename it, remember its name! It will be used later.

11. Double-click the label “0Vdc”. A similar window will appear. Here you assign a DC voltage to the source. Assign 12 V by typing “12” into the box as shown:



Note: Do not type “12V” or any other combination of letters, simply the number 12 will suffice. Once you are done, click OK.

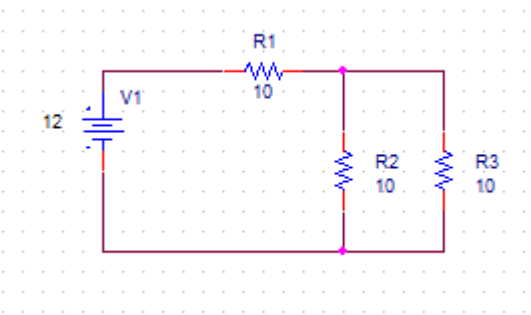
12. Place resistors by typing “r” into the “Part” box. The item “R/ANALOG” will be highlighted, double-click it to place resistors. Place three resistors and reassign each of them to have a resistance of 10 Ω (in exactly the same way we assigned a voltage value to the voltage source in the previous step) as shown:



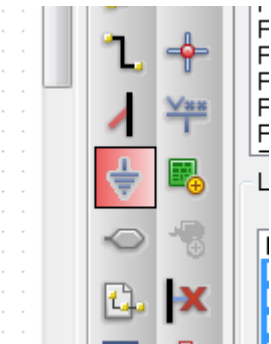
Note: You can rotate a resistor by clicking on it to select it, then pressing “R” on your keyboard while it is selected.

Note: The default value of “1k” with which the resistors were created means 1 kilo ohm, or 1000 Ω . OrCAD will accept any SI postfix to modify numerical values in this manner.

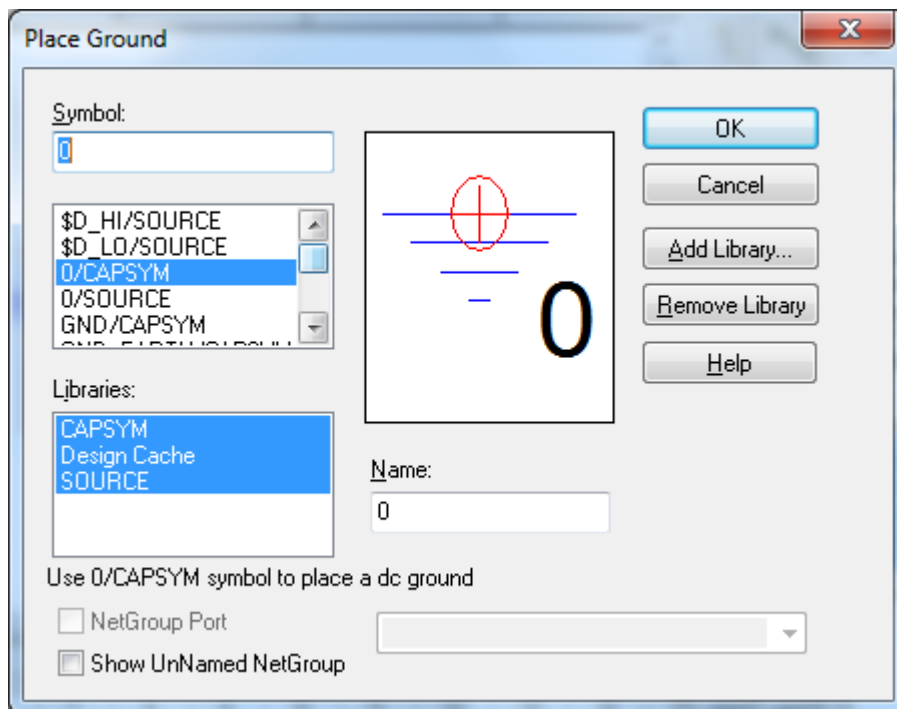
13. Once you have placed the resistors, press “Esc.” on your keyboard to make sure that your cursor is clear, then press “W” to begin placing wires. Place wires by clicking once where the wire should start, then once where it should end. Connect the components you have placed in the following manner:



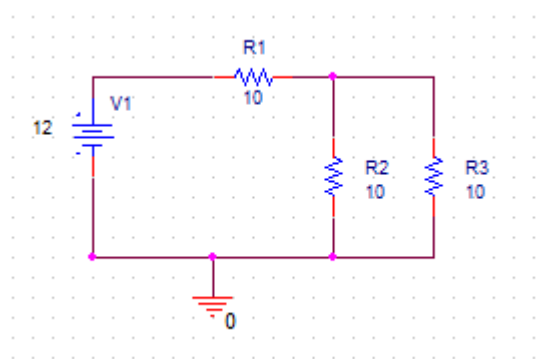
14. Before the circuit can be simulated, a ground reference is required. You can add a ground by clicking on the “Place ground” button to the right of your screen:



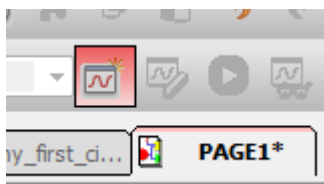
The following window will appear:



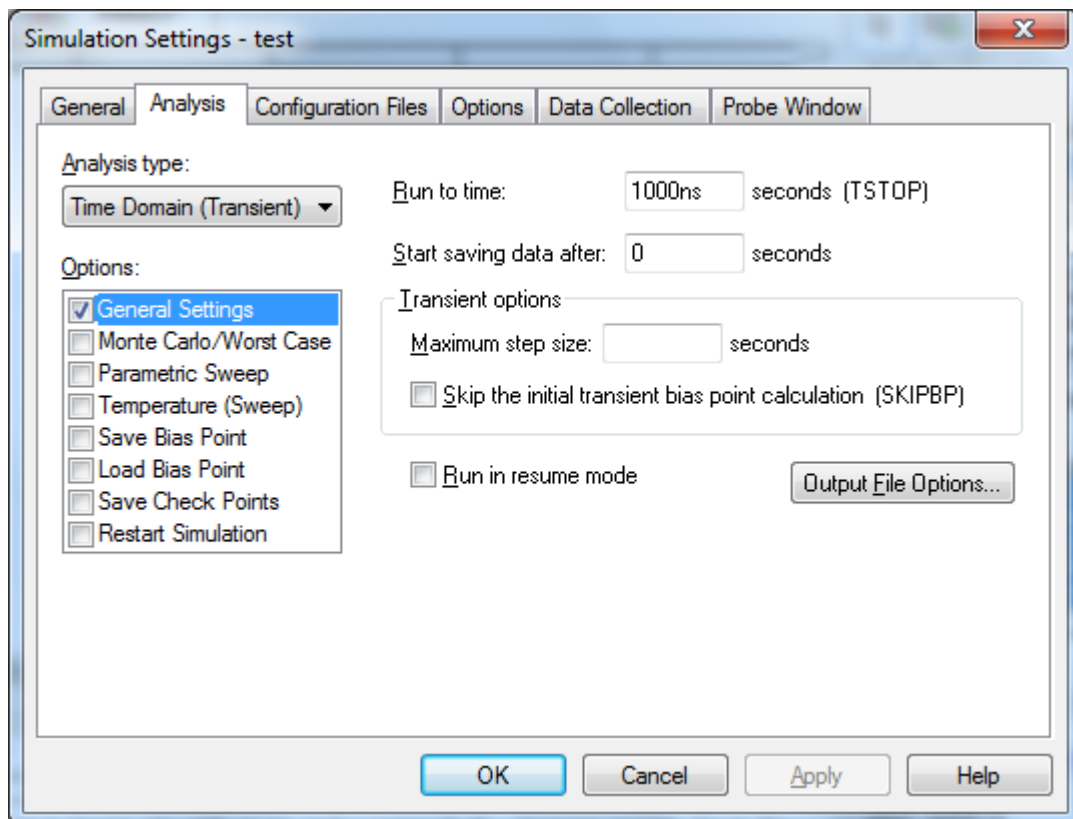
Double-click in “0/SOURCE” and place the ground somewhere appropriate on your schematic. Connect it to the rest of the circuit with a wire as in the previous steps.



15. Your circuit is now ready to simulate! The first step is to create a PSpice simulation profile. The button is located near the top of the window:

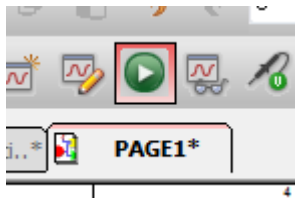


In the window that follows, give the simulation profile any name you feel is appropriate and click “Create”. The following window will appear:



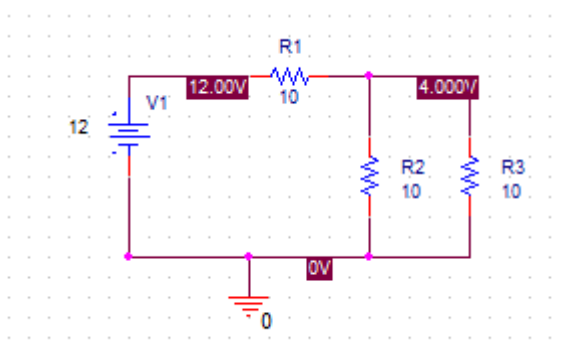
Note: In some instances, the window might appear minimised. You can bring it to the front using the taskbar.

16. Under “Analysis type:” select “Bias Point”. Then click OK.
17. Start the simulation by clicking the ___ button:



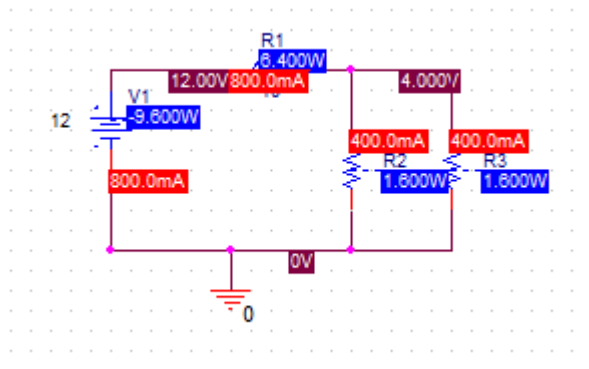
A warning will pop up, informing you that changes will be saved and the undo/redo functionality will not be available after this point. This is normal. Click “Yes” to proceed with the simulation.

18. A PSpice window will appear, but we have not asked PSpice to graph anything, so there will be no graph in it. Rather the DC bias point of your circuit will be shown on your schematic:



By default, only the node voltages (with reference to ground) are shown at each node. You can also show currents in each branch and power dissipated or provided by each circuit

element by clicking on the appropriate buttons.



19. The buttons just to the left of these are for placing probes.



These probes measure node voltage, voltage with a different reference, current and power respectively and are not particularly useful with a bias-point simulation. They are however useful with the other kinds of simulations.

20. In your practical, you will be required to use the “DC Sweep” simulation. Take this opportunity to experiment with it, to see what it does. Try to predict the results of your simulation based on your knowledge of circuit theory before you run the simulation. You may edit the PSpice simulation profile by clicking on the following button:



This will return you to the window seen in step 15.